Miguel Romero

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Education

Pennsylvania State University University Park, PA
Bachelor of Science in Computer Science *Aug. 2022 – May 2026*

Relevant Coursework: Machine Learning and Algorithmic AI, Artificial Intelligence, Exploratory Data Mining, Intro to Cryptography, Database Management Systems

Technical Skills

Languages: Python, C, C++, Java, SQL, Verilog
Tools: Git, Linux, VS Code, IntelliJ, PyCharm, Visual Studio
Core CS: Data Structures, Algorithms, Operating Systems, Computer Organization, Databases, Digital Logic, Assembly Language

Projects

Chess Coach AI | *Python, FastAPI, Stockfish, LLM APIs* 2026

- Built an AI-powered chess coaching platform that analyzes imported game histories, identifies recurring mistakes and opening weaknesses, and explains best moves using engine-based evaluation and natural-language feedback.
- Created an interactive board review workflow with move-by-move analysis, personalized coaching insights, and human-readable explanations generated from structured engine output.
- Integrated Stockfish analysis with LLM-based feedback to translate raw position evaluations into actionable lessons for players.

Multi-Agent UAV Path Planning Simulation | *NVIDIA Isaac Sim, Python, Robotics Simulation* 2026

- Built a multi-agent drone path planning simulation in NVIDIA Isaac Sim to test coordinated UAV navigation in a physics-based environment.
- Implemented planning and control logic for multiple drones to avoid collisions and execute coordinated movement through constrained spaces.
- Used Isaac Sim's robotics and trajectory-planning workflow to prototype and evaluate autonomous drone behavior before real-world deployment.

FPGA-Accelerated Multi-Agent Path Planning and Collision Detection | *FPGA, Verilog/VHDL, Embedded Systems, F*

- Designed a hardware/software co-architecture for multi-agent UAV path planning, using FPGA acceleration for collision detection and trajectory validation.
- Reduced motion-planning bottlenecks by parallelizing collision checks and integrating accelerated planning into a simulation workflow.
- Tested path feasibility and collision avoidance across simulated drone scenarios to validate real-time navigation performance.

Experience

Resident Assistant Aug. 2023 – May 2024
Penn State University *Hazleton, PA*

- Supported 40+ first-year residents through conflict resolution, mentorship, and community-building events.
- Coordinated with housing staff to enforce dorm policies and connect students with campus resources.

Student Food Service Worker Sep. 2024 – May 2025
Penn State University *State College, PA*

- Assisted with meal preparation, service, and clean-up in a fast-paced campus dining environment.
- Maintained food safety, cleanliness, and customer service standards during high-volume shifts.